

JOHN MATHEW PAROCHA

SOFTWARE ENGINEER

ABOUT ME

Driven by a lifelong passion for technology, I specialize in backend engineering with a focus on building scalable architectures and optimizing system performance. By balancing academic excellence with leadership in campus esports initiatives, I have developed a technical expertise and collaborative problem-solving skills ready for real-world challenges.

CONTACT

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Portfolio: <https://machewportfolio.vercel.app/>

Address: Montalban, Rizal

Academic: BSIT | Polytechnic University of the Philippines - Quezon City

DEVELOPMENT EXPERIENCE

Lead Developer & DevOps Engineer (Capstone) | 2026 | *Unified Health Service Ecosystem – Bernardino General Hospital*

- **Microservices & Core EHR Engine:** Orchestrated an 11-module healthcare platform behind a centralized API Gateway, engineering a robust Electronic Health Record (EHR) registry to handle complex patient transactions including OPD/ER/Inpatient classification and automated real-time billing computations.
- **Cross-Module Integrations & Security:** Engineered asynchronous data pipelines to seamlessly append telemedicine consultations and laboratory results to patient records, secured by a custom SSO solution featuring biometric enrollment, 3-layer JWT validation, and granular RBAC.
- **Predictive Analytics & AI Isolation:** Deployed clinical risk predictive models utilizing ClickHouse and pgvector, and integrated a RAG-powered Natural Language Chatbot with strict algorithmic guardrails engineered to isolate and prevent AI querying of sensitive EHR or financial transaction data.

Full-Stack Developer (Freelance) | June 2026 | *Wedding RSVP & Digital Pass Platform*

- **Biometric Authentication:** Engineered a secure, three-phase player onboarding system capturing live face embeddings and ensuring absolute uniqueness across the platform using pgvector cosine similarity matching.
- **Blockchain Integration (Web3):** Integrated EVM-compatible smart contracts using Solidity and web3.py to securely mint and distribute Soulbound Tokens (SBTs) and certificates, gated by live biometric verification and EIP-712 signatures to prevent replay attacks.

Full-Stack Developer (Academic) | March 2026 | *VeriPlay – Biometric & Web 2.5 Tournament Management System*

- **Intelligent Guest Onboarding:** Developed a seamless, animated single-page RSVP platform utilizing Server-Side fuzzy name matching to protect guest privacy while ensuring accurate record lookup.
- **Automated Digital Ticketing:** Built a secure, automated QR-code generation system for digital wedding passes, featuring instant high-resolution downloads and asynchronous email delivery via Resend.

Full-Stack Developer (Freelance) | March 2026 | *TypeCognito – Intelligent Assessment Platform*

- **Multi-Modal Cognitive Assessment:** Built an interactive educational platform using Next.js and Framer Motion, merging standard typing tests with complex narrative analysis and reading comprehension evaluations.
- **Advanced Academic Analytics:** Developed a sophisticated oversight dashboard for administrators featuring real-time Recharts radar visualizations, while securing the assessment loop with rigorous custom event interception to prevent academic dishonesty.

Full-Stack Developer (Freelance) | December 2025 | *Payroll Management System – Multi-Branch Retail Client*

- **Automated Payroll Engine:** Architected a sophisticated calculation system leveraging formula-based logic and individual overrides to manage complex compensation, deductions, and savings tracking across multiple retail locations.
- **Workforce & Security Management:** Integrated real-time timekeeping with strict Role-Based Access Control (RBAC), ensuring secure tracking of employee shifts and high-performance financial audit trails via optimized database indexing.

Full-Stack Developer (Freelance) | October 2025 | *EdVocate (HIV Awareness Platform) Pasig Social Hygiene Clinic*

- **Digital Health Platform:** Engineered a comprehensive web platform to support a formal HIV awareness feasibility study, integrating interactive educational modules and a QR-driven community art submission portal.
- **Geospatial Resource Mapping:** Implemented live map integrations to locate nearby testing clinics and centralized critical healthcare resources, video tutorials, and emergency hotlines.

CERTIFICATIONS

- Programming for Beginners Using Python
- Programming for Intermediate Using Python
- Learn Basic Statistics with Python
- Analyze Data with Python
- Visualize Data with Python
- Build Python Web Apps with Flask
- Principles of Web Development
- Using HTML and CSS to Design a Website
- Basic Javascript for Web Development
- Developing a Static Website
- Digital Transformation 101
- Intermediate Level of Digital Transformation
- Advanced Level of Digital Transformation
- MABL Foundations - QA Automation